2.1.1 Roots of a Quadratic Equation:

Algorithm:

Step 1: Start
Step 2: Input values of a, b and c
Step 3: Calculate the discriminant

$D = b^2 - 4*a*c$

Step 4: if D > 0, then

Calculate two real and different rots

Display root1 and root2

Step 5: Else if D == 0 , then

Calculate one real and equal root

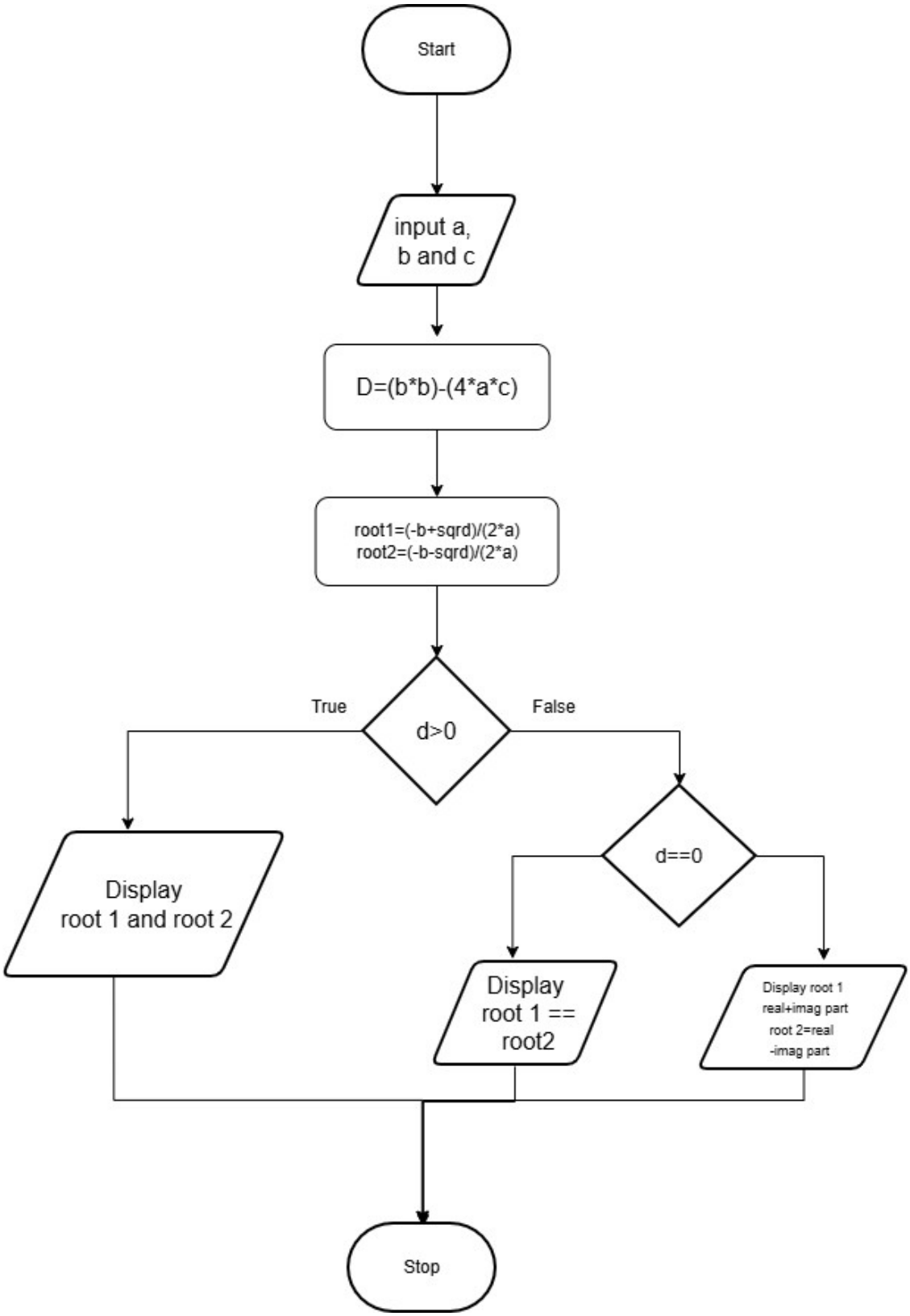
Step 6: Else (D < 0) , then

Calculate two complex roots

Display complex root1 and root2

Step 7: Stop

Flowchart:


Start
input a,
b and c
D=(b*b)-(4*a*c)
root1=(-b+sqrd)/(2*a)
root2=(-b-sqrd)/(2*a)
d>0
True
False
Display
root 1 and root 2
d==0
Display
root 1 ==
root2
Display root 1
real+imag part
root 2=real
-imag part
Stop

2.1.1. Roots of a Quadratic Equation

04:06

coefficients a, b, and c. Use the quadratic formula: $\frac{\left(-b \pm \sqrt{b^2 - 4ac}\right)}{2a}$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a, b, and c, respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = root2 = <Root1>
```

- If roots are imaginary, print:

```
root1 = <RealPart>+<ImaginaryPart>i
root2 = <RealPart>-<ImaginaryPart>i
```

- All values should be formatted to two decimal places.

Sample Test Cases +

Explorer

quadratic...

Submit

```python
a,b,c=map(float,input().split())
d=(b*b)-(4*a*c)
sqrd=d**0.5
root1=(-b+sqrd)/(2*a)
root2=(-b-sqrd)/(2*a)
if(d>0):
    print(f"root1 = {root1:.2f}")
    print(f"root2 = {root2:.2f}")
elif(d==0):
    print(f"root1 = root2 = {root1:.2f}")
else:
    print(f"root1 = {root1.real:.2f}+{root1.imag:.2f}i")
    print(f"root2 = {root2.real:.2f}{root2.imag:.2f}i")
```

Debugger

Terminal Test cases